

4 Ideas for the Packaging Process

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Abstract

There are 4 ideas we came up with which are vacuum pump, orbital wrap, fabric bag with velcro, black sleeve. We are going to explain working principle of each idea and evaluate all ideas to choose the best option.

1. Working Principle

1.1 Vacuum Pump

The process begins with a forklift placing the steel plate onto the infeed conveyor. The conveyor automatically transports the plate into the packaging area. In this section, a pre-folded PA/PE bag is positioned manually by an operator. A new bag must be placed for each cycle. As the plate moves forward on the conveyor, it enters the open bag, which unfolds around it and fully encloses the part. Once the plate is completely inside the bag, it is transferred to the next conveyor section leading to the sealing station. The bag is then sealed using a heat-sealing system, and air is removed to create a vacuum. Finally, the packaged plate exits the machine on the outfeed conveyor, where a forklift picks it up and transports it to storage or shipping.

1.2 Orbital Wrap

First of all, the metal sheets are transported into the orbital wrapping machine using a conveyor. Then the sheet is centered and a rotating ring wraps protective film around it. Once the wrapping is completed, the sheet is automatically released for the next step.

1.3 Fabric Bag with Velcro

First, Metal sheets on the conveyor will be fed into the packaging machine using magnetic grippers and crane. Then the metal sheet will be placed onto the bag and automatically wrapped around using Velcro. Finally, reinforced areas are applied to

each corner and protective housings are installed over the lifting lugs. We can create hole aligned with lugs in the bag.

1.4 Black Sleeve

The sheet element is first placed on supports and is put inside a strong black sleeve. The sleeve covers the whole product and protects it from sunlight, water, and dirt. After that, the open end of the sleeve is closed using straps or a flap. Inside the sleeve, there are small bags(desiccant bag) that absorb moisture to stop rust from forming. The sleeve is tight and strong. So it does not tear easily during transport. When the product arrives, the sleeve can be opened and removed quickly.

2. Evaluation Criteria

	Airtight	UV Protection	Handle Size Variation	Safety	Serial Production	Low Capital Cost	No Marks	Mean
Weight	5	5	3	5	3	2	5	N/A
Vacuum	5.0/5	5.0/5	3.7/5	5.0/5	4.5/5	4.3/5	5.0/5	4.757
Orbital	4.0/5	5.0/5	4.5/5	5.0/5	5.0/5	4.7/5	5.0/5	4.746
Fabric	1.5/5	5.0/5	3.5/5	4.5/5	3.5/5	4.7/5	4.5/5	3.786
Sleeve	2.5/5	5.0/5	4.0/5	4.5/5	3.0/5	4.8/5	4.7/5	4.039

We evaluated our four ideas using seven criteria (airtightness, UV protection, etc.) and assigned a weight to each criterion based on its importance.

2.1 Airtightness

Orbital wrapping received a slightly lower score than vacuum packaging because, although it provides strong protective wrapping, it does not create a completely sealed environment. Small air gaps or incomplete sealing may still allow limited moisture or air penetration, making it less effective for corrosion prevention than vacuum packaging. Fabric bag packaging received a significantly lower score because Velcro closures and fabric materials cannot fully isolate the product from external air or moisture. This limited sealing performance reduces protection against humidity and corrosion. Black sleeve packaging also received a relatively low score because, despite

offering external coverage, it does not provide true airtight sealing. Moisture and air can still enter through openings or loose areas, making it less suitable for long-term rust prevention.

2.2 UV Protection

All four packaging concepts received full scores in UV protection because each design fully covers the metal sheet using protective materials that effectively block direct sunlight. Whether through vacuum film, orbital wrap, fabric covering, or black sleeve material, all solutions provide sufficient protection against UV exposure.

2.3 Handle Size Variation

Orbital wrapping received the highest score because its automated wrapping system can efficiently adapt to various sheet sizes with minimal structural changes. Black sleeve packaging also scored highly due to its flexible material, which can cover different dimensions relatively easily. Fabric bag packaging offered moderate adaptability, but required more customization such as Velcro fitting and lifting lug alignment, reducing flexibility. Vacuum packaging received the lowest score in this category because vacuum chambers or sealing systems often require more precise size adjustments for different products, limiting versatility.

2.4 Safety

Vacuum packaging and orbital wrapping both received the highest scores because their highly automated processes significantly reduce human intervention, lowering operational risks. Fabric bag and black sleeve concepts scored slightly lower because they involve more manual handling steps, such as fitting, fastening, or positioning, which can increase worker exposure to potential hazards. Although these methods are still relatively safe, reduced automation creates more opportunities for human error or handling-related accidents.

2.5 Serial Production

Orbital wrapping achieved the highest score because it is particularly suitable for continuous, high-speed industrial packaging operations. Vacuum packaging also performed strongly, but slightly lower due to potentially slower sealing cycles. Fabric bag and black sleeve concepts scored lower because they require more complex or slower packaging processes, reducing efficiency for large-scale serial production. Manual or semi-automated elements in these designs limit throughput compared to

dedicated industrial machinery.

2.6 Low Capital Cost

Black sleeve and fabric bag concepts received the highest scores because they require relatively simple materials and minimal machinery, resulting in lower initial investment costs. Orbital wrapping also performed well, as although machinery is required, the system is less complex than vacuum infrastructure. Vacuum packaging received the lowest score because specialized vacuum pumps, sealing systems, and automation equipment significantly increase startup costs.

2.7 No Marks

Vacuum packaging and orbital wrapping both received the highest scores because they secure the product tightly, minimizing movement and reducing the risk of scratches or surface marks during transportation. Black sleeve packaging scored slightly lower because some internal movement may still occur, potentially causing friction damage. Fabric bag packaging also performed reasonably well, but fitting complexity and material movement created a slightly greater chance of surface imperfections.

3. The Final Idea

Although both vacuum packaging and orbital wrapping achieved similarly high evaluation scores, vacuum packaging was ultimately selected as the optimal solution. This decision was primarily based on its superior airtightness, enhanced moisture prevention, and stronger long-term preservation performance, which are critical factors for protecting sheet metal products during storage and transportation. Therefore, we decided to proceed with the vacuum idea.